Transforming passive planetarium lectures into interactive, student-driven experiences — grounded in educator insights and enabled by augmented reality

Interviews:

Before designing e*xtrasolAR*, we interviewed 5 exoplanet educators about how + what they teach about exoplanets and what they want from an exoplanet lab.

The Present:

Use of Planetariums for Exoplanets

- showing all known exoplanets
- flying to exoplanets
- biases in which exoplanets we've discovered
- don't use the planetarium because:
  - lacks interactivity
  - lacks meaningful material
  - lack of independence in the dome

Current Lab and Criticisms

- a Lego orrery with a light meter
- only explains the transit method
- not the true scale of real-life data collection
- simplifies too much
- doesn't show the biases of each detection method

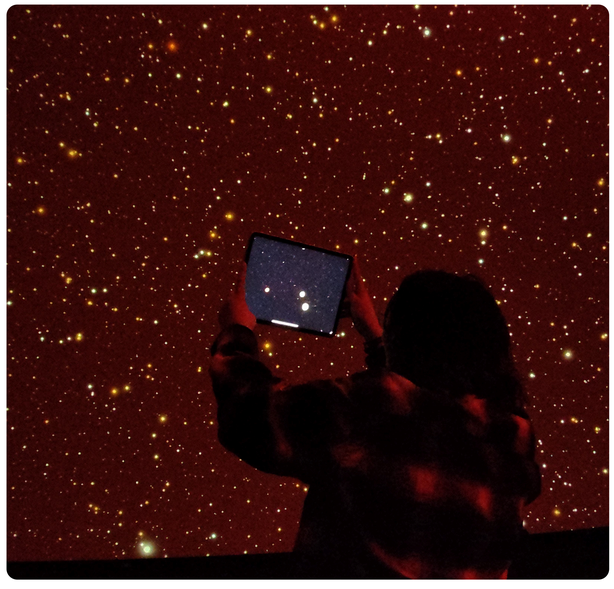
Requirements for an AR Exoplanet Lab:

Their Design Goals

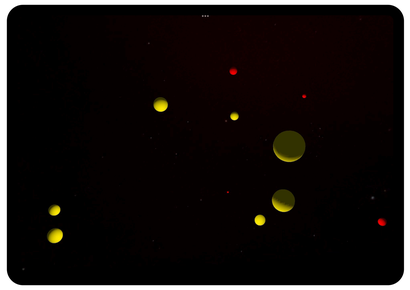
1. scaffolding for students
2. use real data
3. animation to explain what data represents

Their Learning Goals

4. Exoplanet Detection Methods
5. Properties and Biases of the Detection Methods
6. Properties' Implications on Life
7. Plurality of Exoplanets

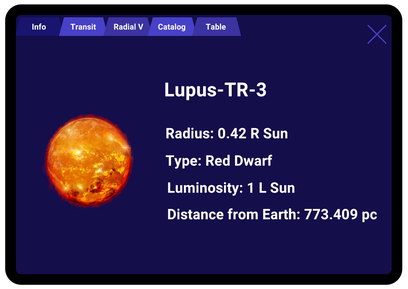


extrasolAR:



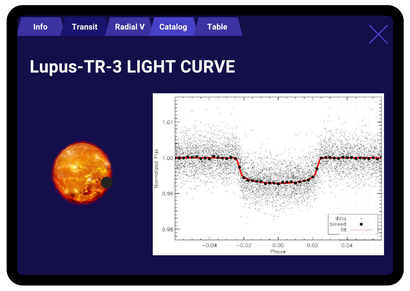
Stars Display

(7)



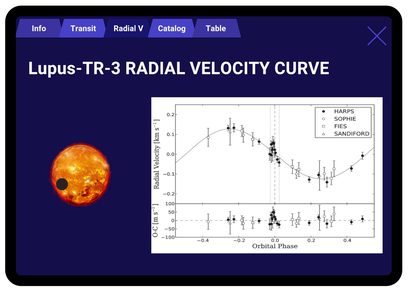
Star Information

(2)



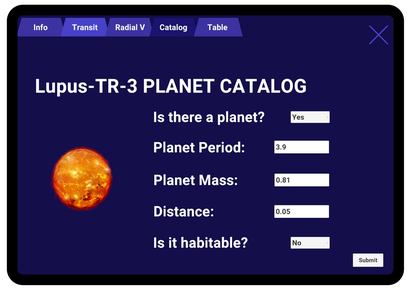
Transit Data

(2) (3) (4)



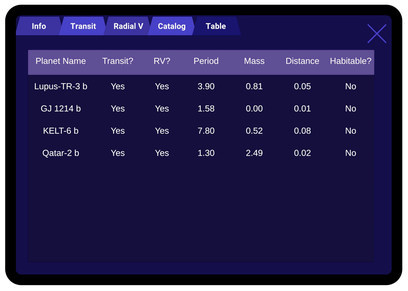
Radial Velocity Data

(2) (3) (4)



Exoplanet Catalog

(1) (5) (6)



Exoplanets Table

(5) (7)

Designing an augmented reality exoplanet lab app and activity to facilitate interaction and active learning in the planetarium

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